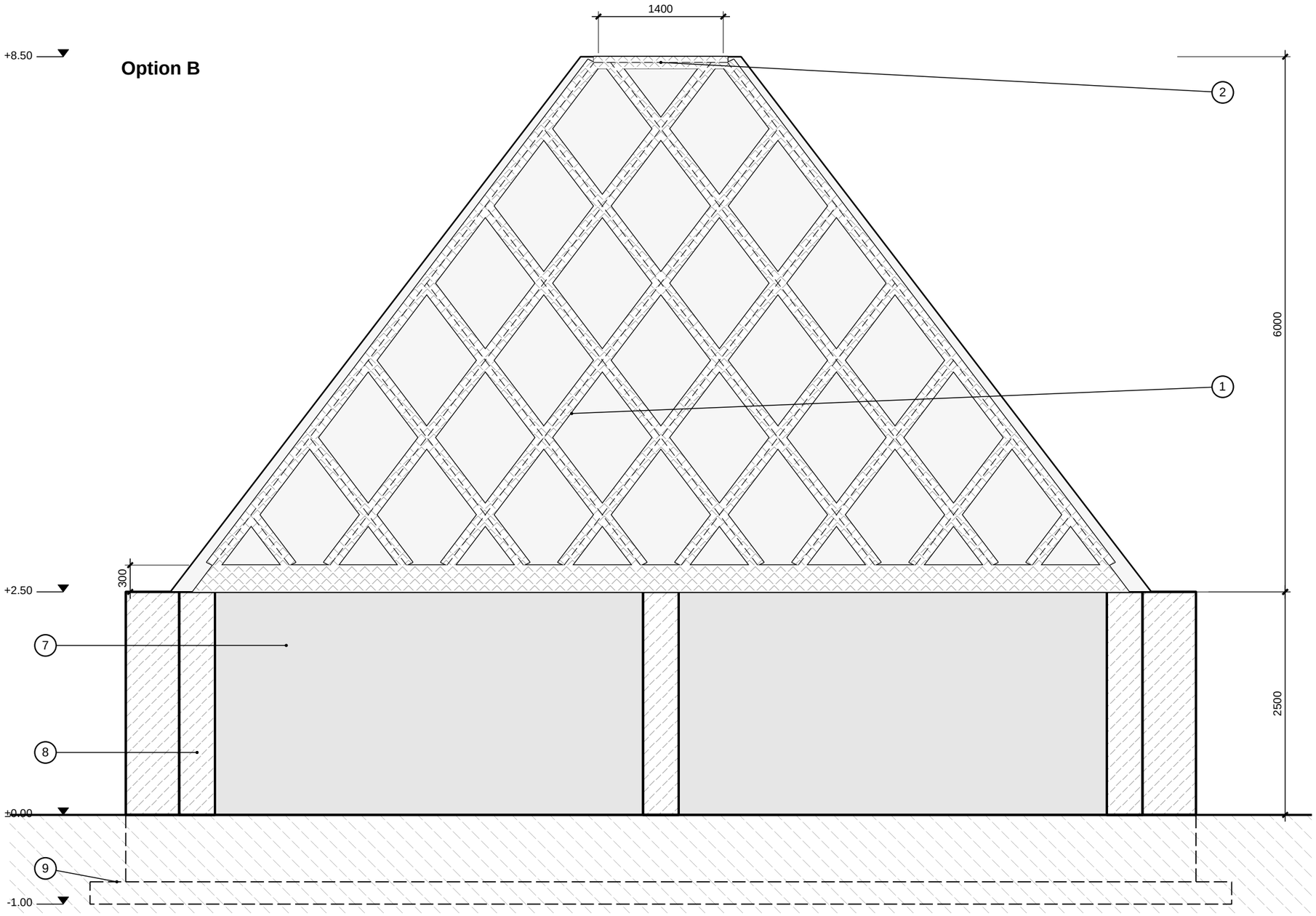
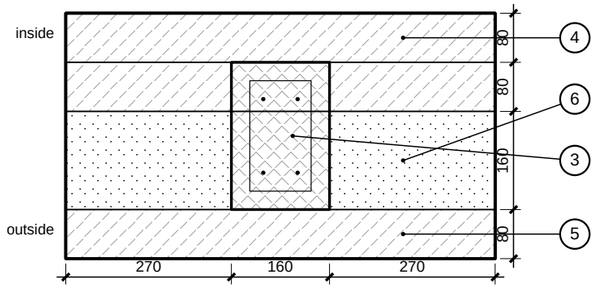


Section through one rib, square to the slope, 1:10



Roof grid B: diamond lattice

one roof face seen square on

(1) Diagonals parallel to the two hips (same inclination as the roof edges, 58.9° in the face), every 1312.5 along the base: equal diamonds 1312.5 wide.

(2) Bottom ring = the 300 solid base; top ring at the crown. The hips are two of the diagonals. No other horizontal ribs.

(3*) Rib (column): printed channel 160 wide x 240 deep in the 160 air zone + layer 2, filled with cast concrete, 4 Ø16 + Ø8 stirrups every 200 (assumed).

(4) Inner layer 1, 80: continuous, covers the grid on the room side.

(5) Outer skin 80: continuous, covers the grid on the outside (cladding on it).

(6*) Between the ribs the air zone stays open for thermal insulation (dotted).

(7) Printed wall 400 (same layers as the roof), to +2.50.

(8) Buttresses 400 wide, 1000 from the inner face (600 outside), at the corners and mid-span; roof ribs and wall bars tie into the solid base above them.

(9*) Strip footing 1800 x 250, underside at -1.00 (hidden, dashed).

* = assumed, to confirm

Notes

Every diagonal ends on a ring or on a hip, so no bar has a free end in the shell.

Diagonals do not stop as the roof narrows: the diamonds just get clipped by the hips.

Rough quantities, four faces (hips counted twice, so slightly high): ribs 330 m, 4 Ø16 each bars, steel 2085 kg, cast concrete 12.7 m3.

Rib size, bar count and the 160 x 240 channel are assumptions for the engineer to check. Elevation shows the face square on; true lengths are 1.25 x the drawn heights.

0 1 2 3 4 5 m

Scale 1:50 at A3 · dimensions in mm · draft 1, 2026-10-11
pyramid-roof · hand estimates, not engineered